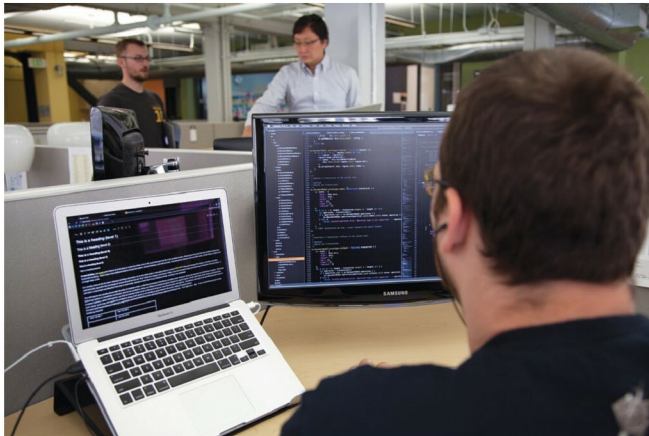


Property-based Testing in Haskell

and How to Install Cabal



Haskell, named after logician Haskell Curry, is a standardised, general-purpose, purely functional programming language, with non-strict semantics and strong static typing. Let's take a look at the property-based testing of Haskell programs and at the Cabal tool, which is used to build and manage Haskell packages and applications.

One of the main features of testing in Haskell is property-based testing. The type system allows you to infer and derive types, and also helps in auto-generating test cases. QuickCheck is a popular property-based testing library for Haskell. If your program is pure, you can write tests to ascertain the properties and invariants of your programs, and the tests can be auto-generated and executed.

You can install QuickCheck on Fedora, for example, by using the following command:

```
$ sudo yum install ghc-QuickCheck-devel
```

Consider a simple function to add two integers:

```
mySum :: Int -> Int -> Int
```

```
mySum a b = a + b
```

You can ascertain the property of the function that 'a + b' is the same as 'b + a' by using the QuickCheck library. You must first define the invariant in a function as shown below:

```
prop_mySum a b = mySum a b == mySum b a
```

You can test the code directly in the GHCi prompt, using the following command:

```
$ ghci sum.hs
GHCi, version 7.6.3: http://www.haskell.org/ghci/ :? for help
Loading package ghc-prim ... linking ... done.
Loading package integer-gmp ... linking ... done.
```